Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

Sound Localization Robot, Team 5

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Lucas Kettler, Joshua Teele

TA: Fahrettin Ay
Sponsor: Dr. Moreira

Project Scope

Figure 1. First responder crawls through smoke and broken glass to rescue elderly man from house fire (Independent TV)

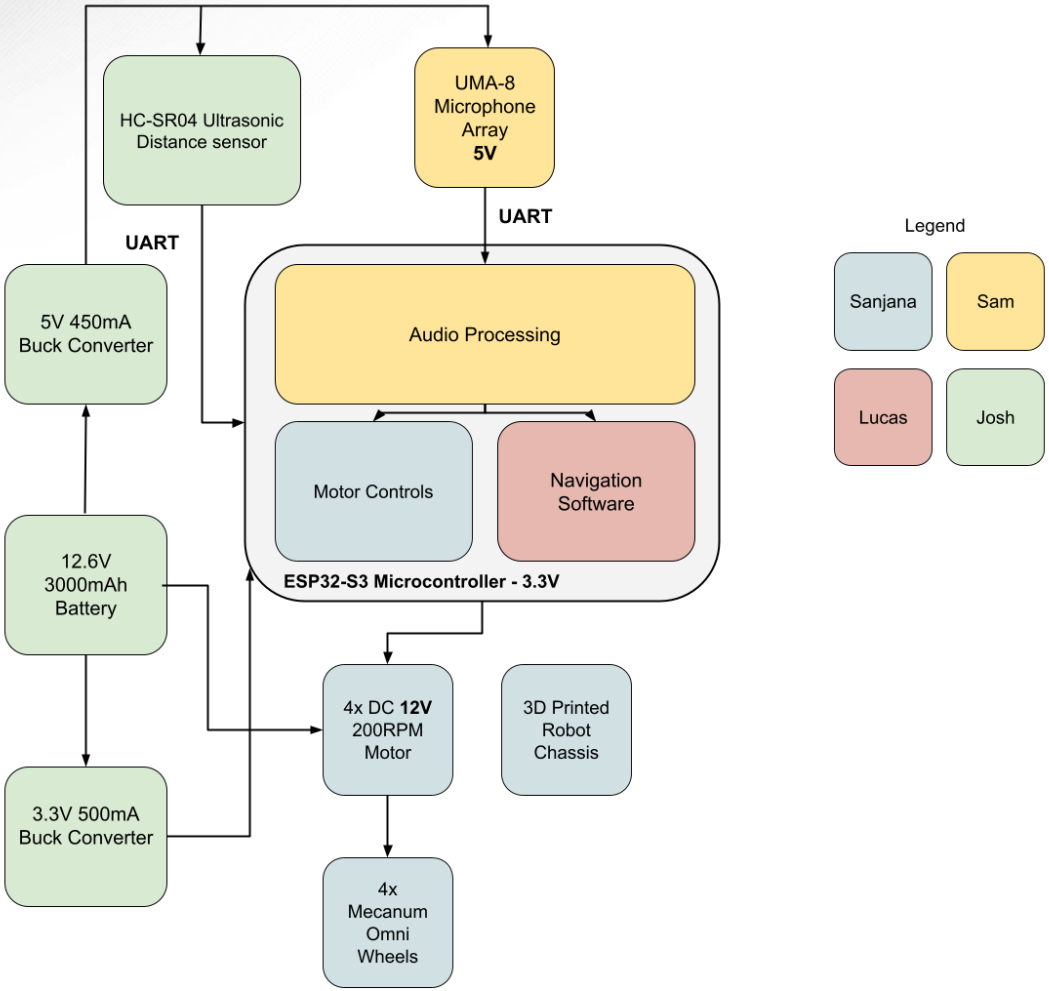
- Keeps **first responders safe** by entering hazardous or toxic areas first.
- **Reduces search and rescue time** by avoiding exploring unnecessary paths
- **Reliable in low-visibility** or high-stress conditions (smoke, debris, noise).

Problem: First Responders lack reliable methods for navigating dangerous environments, where visibility is limited, putting both their safety and the victims at risk.

Our Solution: Building a Sound Localization robot that can hear distress calls, pinpoint their direction, and move toward them autonomously, allowing first responders to focus on saving lives instead of risking theirs.

*A smarter, safer, and faster way to locate victims, all while protecting rescuers and saving more lives.

Robot Sub Systems:



Input Signal Software

Samuel Lackey

Accomplishments made since last presentation:	Ongoing plans for final Demonstration
• Found sound can still be detected with obstacles • Added a rough distance calculation	• Create a webapp to view live data from the ESP32

Accomplishments made in 403:
• TDOA algorithm working within 3.6° accuracy • Data outputs from mic Array and onto ESP32 • Algorithm running on ESP 32

Input Signal Software

Samuel Lackey

```
// --- Bandpass Filter & Gain ---
const float MIN_FREQ = 300.0f;
const float MAX_FREQ = 3400.0f;
const float VOICE_FREQ_GAIN = 10.0f;

// --- DISTANCE ESTIMATION (NEEDS CALIBRATION!) ---
const float REFERENCE_DISTANCE = 0.16f;
const float REFERENCE_ENERGY = 0.05f;
```

Figure 1. Calibration and filter constants

```
========================================
Peak Frame RMS:        0.0029 (used for distance)
Detected Angle:        0.9 degrees
Estimated Distance:    ~0.66 meters
========================================
```

Figure 2. Time Delay with Distance

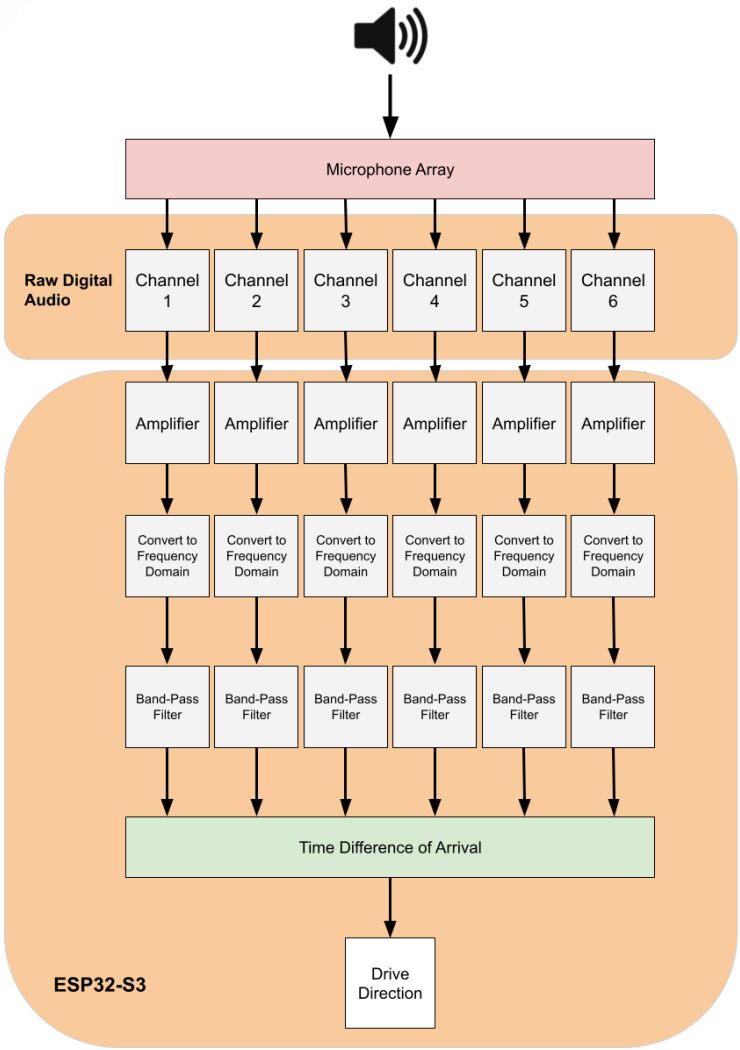


Figure 3. Subsystem Diagram

Navigation Software Subsystem

Lucas Kettler

Accomplishments made since last presentation:	Ongoing plans for final Demonstration:
• Ultrasonic sensor (HC-SR04) implementation, and algorithm testing in Wokwi simulation • Modification of Algorithm to handle edge cases	• Document multiple trials runs per simulation scenario • Perform the simulation test cases with the modified Algorithm using and the completed sound localization robot

Accomplishments made in 403:

- Implementation and testing of A* algorithm in C++ with a digital grid map
- Added a map expansion function
- Enabled use of a position shift function of goal and starting node in edge cases
- Performed hardware simulations with sensor input

Navigation Software

Lucas Kettler

```
// ==================== ULTRASONIC SENSOR SIMULATION ============
// This version fakes distance readings by generating AC-like dis
// occasionally injecting a "close object" reading. Real mode rep

float getDistanceCM(int trig, int echo) {
  // Ultra-short HIGH pulse (as if triggering a sensor)
  digitalWrite(trig, LOW);  delayMicroseconds(2);
  digitalWrite(trig, HIGH); delayMicroseconds(10);
  digitalWrite(trig, LOW);

  // Fake distance: between 5cm and ~120cm
  int simulatedCM = random(5, 120);

  // Simulate echo-pulse duration
  unsigned long echoTime = (unsigned long)(simulatedCM * 58.2);

  // Fake echo pulse HIGH for that duration, then LOW again
  // This is for simulation — real mode uses pulseIn()
  pinMode(echo, OUTPUT);
  digitalWrite(echo, HIGH);
  delayMicroseconds(echoTime);
  digitalWrite(echo, LOW);
  pinMode(echo, INPUT);

  return simulatedCM;
}
```

Figure 1. Simulated Sensor Input

```
// grid[y][x]: 0 = free space, 1 = obstacle
std::vector<std::vector<int>> grid = {
    {0, 0, 0, 0, 1},
    {1, 1, 0, 1, 0},
    {0, 0, 0, 0, 0},
    {0, 1, 1, 1, 0},
    {0, 0, 1, 0, 1}
};
```

Figure 2. Expanding grid map

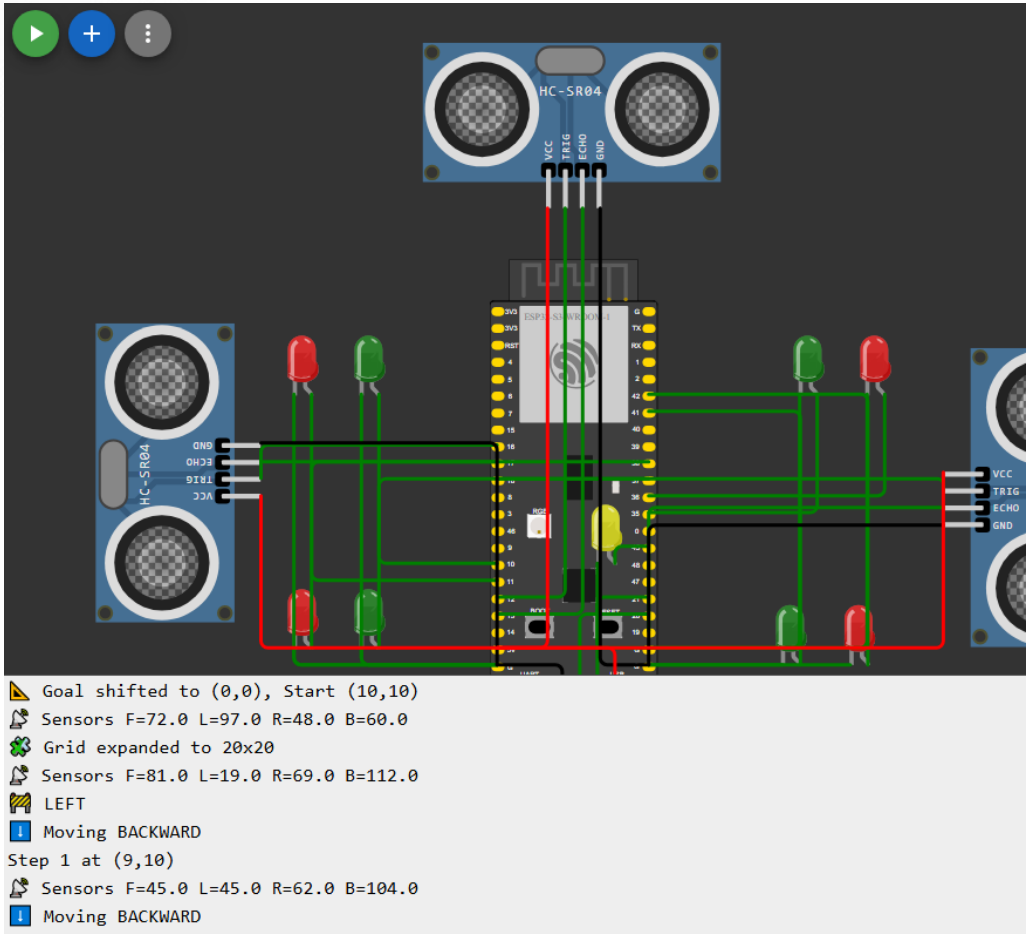


Figure 3. Wokwi Simulation

Mechanical/ Motor Controls Subsystem

Sanjana Pasumarthi

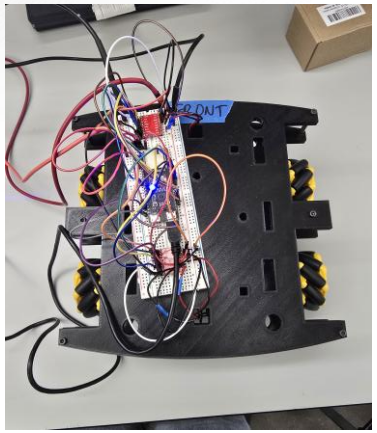
Accomplishments made since last presentation:	Ongoing plans for final Demonstration:
• Soldered some permanent wiring connections • Debugged code and confirmed pins functionality and signals • Firmware and software validation for bidirectional, lateral, and stalling movement • Assembly of Chassis	• Robot can take in user commands and execute without any errors • Maintain remote connection between ESP32 and user interface without any interruptions • Solder motor drivers & ESP32 together

Accomplishments made in 403:
• 3D printed robot chassis to physically hold components together • Soldered wiring connections for certain components • Programmed ESP32 to control motor drivers/ motors for movement • Built testbench to test the signals sent to motor drivers

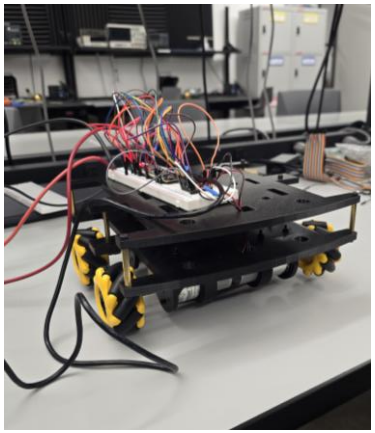
Mechanical/ Motor Controls Subsystem

PWM Test

Top View



Cross View



- Motors move in all 4 directions: Forward Backward, Left and Right
- Wires need to be soldered for final demonstration

- PWM Frequency: 1 kHz, displayed from all 4 PWM pins at input of motor driver
- Step Function: confirms ESP32 is generating on/off pulses not just some digital writes high/ low
- Helped in Debugging to make sure GPIO functions & checks signal quality

Testbench Signal Verification/ Software

```
#define MF_PWMA 4
#define MF_AI1 8
#define MF_AI2 18
```

```
// Set all connections as outputs
pinMode(MF_PWMA, OUTPUT);
pinMode(MF_AI1, OUTPUT);
pinMode(MF_AI2, OUTPUT);
delayMicroseconds(10);
```

```
MF_PWMB: 200 MF_BI1: 0 MF_BI2: 1 | MR_PWMA: 200 MR_AI1: 0 MR_AI2: 1 | MR_PWMB: 200 MR_BI1: 1 MR_BI2: 0
stopMotors | MF_PWMA: 200 MF_AI1: 0 MF_AI2: 0 | MF_PWMB: 200 MF_BI1: 0 MF_BI2: 0 | MR_PWMA: 200 MR_AI1: 0 MR_AI2: 0 | MR_PWMB:
Straight Forward
moveMotors | MF_PWMA: 200 MF_AI1: 1 MF_AI2: 0 | MF_PWMB: 200 MF_BI1: 1 MF_BI2: 0 | MR_PWMA: 200 MR_AI1: 1 MR_AI2: 0 | MR_PWMB:
stopMotors | MF_PWMA: 200 MF_AI1: 0 MF_AI2: 0 | MF_PWMB: 200 MF_BI1: 0 MF_BI2: 0 | MR_PWMA: 200 MR_AI1: 0 MR_AI2: 0 | MR_PWMB:
Straight Backward
moveMotors | MF_PWMA: 200 MF_AI1: 0 MF_AI2: 1 | MF_PWMB: 200 MF_BI1: 0 MF_BI2: 1 | MR_PWMA: 200 MR_AI1: 0 MR_AI2: 1 | MR_PWMB:
stopMotors | MF_PWMA: 200 MF_AI1: 0 MF_AI2: 0 | MF_PWMB: 200 MF_BI1: 0 MF_BI2: 0 | MR_PWMA: 200 MR_AI1: 0 MR_AI2: 0 | MR_PWMB:
Sideways Right
moveMotors | MF_PWMA: 200 MF_AI1: 0 MF_AI2: 1 | MF_PWMB: 200 MF_BI1: 1 MF_BI2: 0 | MR_PWMA: 200 MR_AI1: 1 MR_AI2: 0 | MR_PWMB:
stopMotors | MF_PWMA: 200 MF_AI1: 0 MF_AI2: 0 | MF_PWMB: 200 MF_BI1: 0 MF_BI2: 0 | MR_PWMA: 200 MR_AI1: 0 MR_AI2: 0 | MR_PWMB:
Sideways Left
moveMotors | MF_PWMA: 200 MF_AI1: 1 MF_AI2: 0 | MF_PWMB: 200 MF_BI1: 0 MF_BI2: 1 | MR_PWMA: 200 MR_AI1: 0 MR_AI2: 1 | MR_PWMB:
stopMotors | MF_PWMA: 200 MF_AI1: 0 MF_AI2: 0 | MF_PWMB: 200 MF_BI1: 0 MF_BI2: 0 | MR_PWMA: 200 MR_AI1: 0 MR_AI2: 0 | MR_PWMB:
```

*Viewed input going into motor drivers matched expected bit control values

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Power & Distance Sensor Subsystem

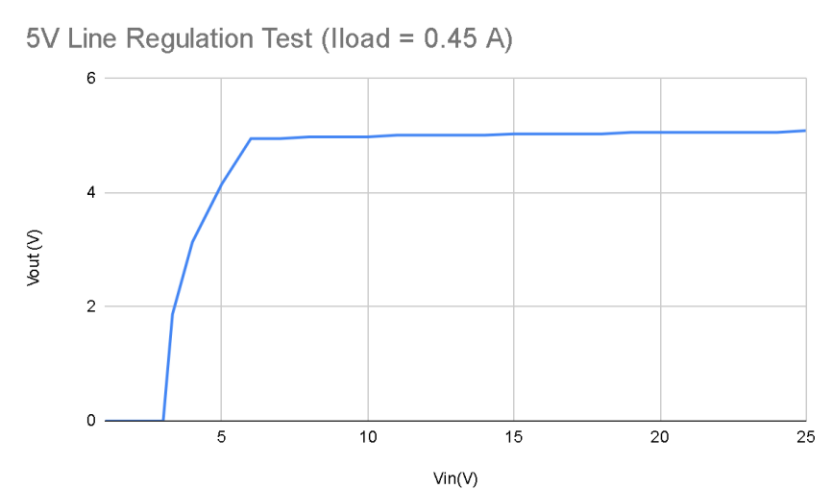
Joshua Teele

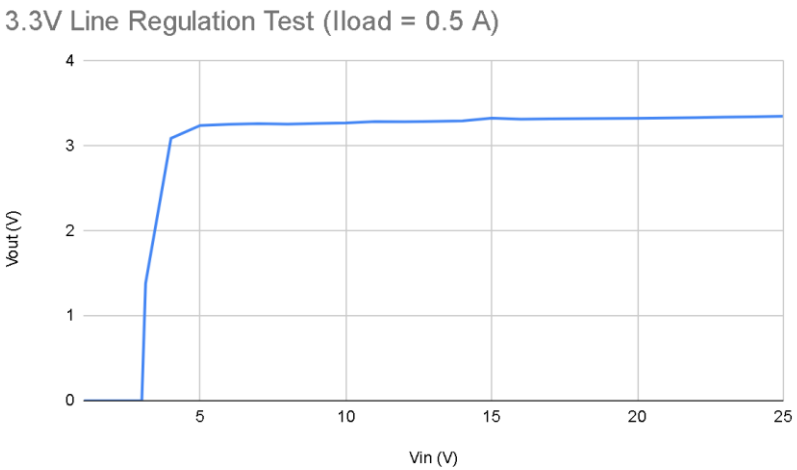
Accomplishments made since last presentation:	Ongoing plans for final presentation
• Performed line and load testing on buck converters that will be used for testing • Soldered system where battery can connect to converters and motors • Created test code for distance sensors	•Put together and test buck converters that I designed •Make the system that battery is connected to look nice and not be as fragile •Create small system with distance sensors

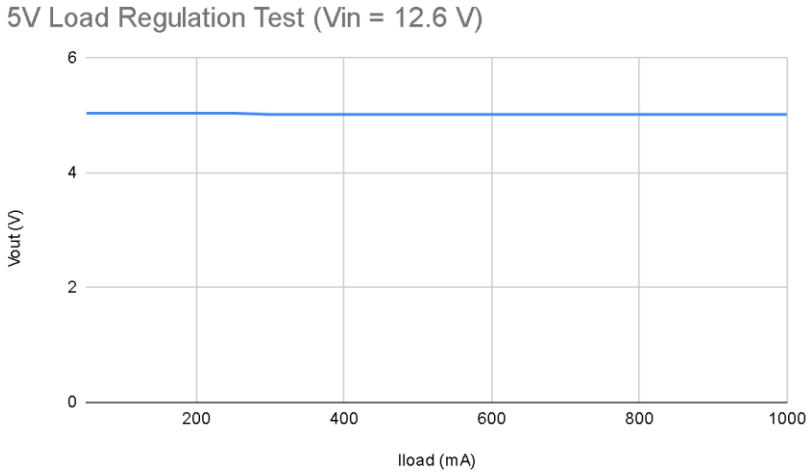
Accomplishments made in 403:
• Designed buck converters to be used to power components on the robot • Created code to run distance sensors

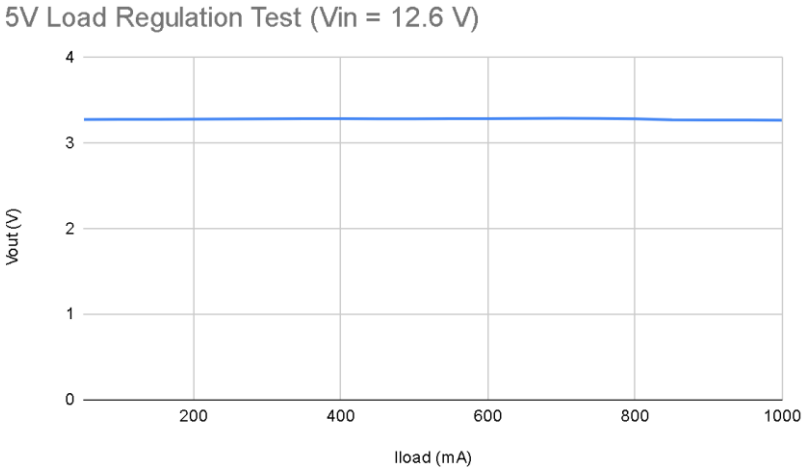
Power & Distance Sensor Subsystem

Joshua Teele









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Validation Plan

Navigation Software Subsystem - Lucas

Test	Objective	Component Tested	Method	Pass Criteria	✓
Ideal Path Functionality	Verify shortest path is taken	Navigation Loop	Run simulation with no obstacles, calculate Manhattan distance, visually verify	Manhattan distance path is taken by the "robot"	✓
Obstacle Detection	Detect obstacles within 9 inches on all sides	Sensor Software	Run a simulation with random ultrasonic sensor input for obstacles	On display obstacles are marked when sensors receive a distance < 9 inches	✓
Obstacle Avoidance	Verify a halt to movement when encountering an obstacle	Motor Control	Run simulation and verify behavior after obstacle detection	The path in the simulation does not continue through an obstacle node	✓
Path replanning	Verify that new path is made after obstacles found	Navigation Loop	After obstacle detection, calculate a new Manhattan distance	The new Manhattan distance path is taken by the "robot"	✓
Expanding Grid map	Apply grid map expansion when given an out-of-bounds goal	Navigation Loop	Run simulation with goal node outside of original grid map	Grid map is expanded to be larger than that of goal node location	✓
Negative direction movement	Handle movements in the [-X,-Y] region	Navigation Loop	Run a simulation with goal node in [-X,-Y] region	Goal node and starting node are shifted to be within bounds	✓
Physical Hardware - I	Validate movement without obstacles	Navigation Software	Real world test environment with goal node hard coded	Goal reached with Manhattan distance path taken	
Physical Hardware - II	Validate movement with obstacles/edge cases	Navigation Software	Real world test with physical obstacles in place	Goal reached for edge cases without making contact with obstacles	

Motor Controls Subsystem - Sanjana

Test	Objective	Component Tested	Method	Pass Criteria	✓
Bidirectional Motor Function	All motors shall operate correctly in both forward and reverse directions	Motors, ESP32-S3, Motor Drivers, All subsystem wiring	All Motors move forward and backward at constant speeds	Test PWM at driver input, ± 5 % of target frequency	✓
Lateral (Strafing) Motion Test	All motors shall coordinate to produce lateral (left and right) translation	Motors, ESP32-S3, Motor Drivers, all subsystem wiring	All Motors move laterally at constant speeds	Motors move backward, test PWM at driver input, ± 5 % of target frequency	✓
Idle State Command Readiness	Remain in an idle state until a valid motion command is received	Motors, ESP32-S3, Motor Drivers	Robot stays in idle state, when no command is given	No PWM signal (0) at driver input	✓
Wireless Connectivity & Stability	Maintain continuous and reliable communication between the ESP32 and the user interface	ESP32-S3	Interface connected to ESP through Wifi or Bluetooth address	Robot connected to interface for at least 10 minutes without interruption	
Remote Control Interface	Capable of receiving and executing control commands via both Bluetooth or Wi-Fi communication interfaces.	ESP32-S3	Robot follows command given by user, and prompts response	Robot executes proper response to command	
Firmware Integration and Execution	Firmware shall compile, upload, and execute without errors	ESP32-S3	Create Testbench that prints signals sent to motor drivers for each function	Testbench displays proper bit values according to direction	✓

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Validation Plan

Power & Distance Sensors Subsystem - Josh

Test	Objective	Component Tested	Method	Pass Criteria	✓
Input Voltage Range Test	Steady output voltage & current for range of input voltages (important range: 11-13V)	Buck converters (from FEDC cage)	Line regulation test using eload	Output voltages stay steady while pulling a constant current	✓
Constant Output Voltage for Changing Load Current	Output voltage must stay constant if the output current changes (±5%)	Buck converters (from FEDC cage)	Load regulation test using eload	Output voltages stay steady while pulling a range of currents	✓
Test Designed PCBs	PCBs output the voltages and currents designed for	Buck converters (designed by me)	Perform line and load regulation tests	Output needed voltages and currents	
Simulate Distance Sensors	Distance sensors output a "stop" signal when detecting something within a small range (5 inches)	Ultrasonic Distance Sensors	Use online simulation software (tinkercad) to simulate code	Sensor outputs a signal which can be read to stop the car (lights up a red LED)	
System wiring and powering	Create wiring system	All involved in wiring system	After putting together, test all system with all components connected	Everything is able to stay on	

Input Signal Software - Sam

Test	Objective	Component Tested	Method	Pass Criteria	✓
I2S Data Integrity	Verify that non-static, valid audio data is being read from the I2S hardware.	UMA-8 Microphone Array ESP32	Place a speaker playing a 1kHz tone then a 100Hz tone.	The 1kHz tone must produce a larger band energy.	✓
DOA Angle Accuracy	Verify the angle calculation is accurate and repeatable for a known, static sound source	UMA-8 Microphone Array	Place speaker playing 1kHz tone at a known angle, and 1m distance. Run the detection test 5 times	All 5 tests must log a detected angle within an 3.6°	✓
Distance Estimation	Verify the calibrated distance logic responds correctly to changes in distance	UMA-8 Microphone Array	Run the distance calibration software, move sound source from 0.5 to 1m to 2m.	The estimated distance must be less than 0.5m of the true distance	✓
Webapp for output data	View live data from the algorithm via a wireless webapp	UMA-8 Microphone Array ESP32	Run the program and connect to wifi, trigger a sound, and access the webapp	The webapp must show the angle, and estimated distance	

Execution Plan

Tests	09/08/25	09/15/25	09/22/25	09/29/25	10/06/25	10/13/25	10/20/25	10/27/25	11/03/25	11/10/25	11/17/2025	11/24/2025	12/01/2025
Motor Controls Subsystem ~ Sanjana Pasumarthi													
Build & Design Chassis	Completed												
Print & Assemble 3D Chassis					Completed								
Firmware Integration & Execution Validation						Completed							
Bidirectional Motor Direction Validation								Completed					
Lateral Motor Direction Validation									Completed				
Remote Control Interface Validation & User Command Testing										In progress			
Navigation Software Subsystem ~ Lucas Kettler													
Search Algorithm	Completed												
Expanding grid Map				Completed									
Obstacle Detection/Avoidance					Completed								
Wokwi Simulations with sensors							Completed						
Edge Case Validation							Completed						
Completed Robot Test								In Progress					
Power & Distance Sensor Subsystem ~ Joshua Teele													
Power Management Creation		Completed											
Design Buck Converter Circuit and PCBs				Completed									
Put together PCBs													
Wire Components and Battery								In Progress					
Create Code For Distance Sensors									In Progress				
Input Signal Subsystem ~ Samuel Lackey													
Funtioning test of microphone		Completed											
TDOA Arrival Algorithm				Completed									
Bandpass Filter							Completed						
Estimated distance								Completed					
Process algorithm on ESP32									Completed				
Output webapp from ESP32											In Progress		

404 Continuation….

Samuel Lackey	Joshua Teele
• Improve TDOA to listen to periodic sounds • DOA should work with obstacles potentially blocking sound • Integrate ESP-32 software to navigation system and motor controls	• Create PCB to connect all components too • Integrate distance sensors into the main functionality of the robot
Lucas Kettler	**Sanjana Pasumarthi**
• Modify Navigation loop to reach out of sight goals given only a direction • Avoid unusually shaped obstacles/terrain	• Design & 3D print physical mounting poles that will hold microphone array in place • Integrate ESP32-S3 software with navigation & input signal system (take in input from other software)

404 Expectations and Mutual Understandings
• Robot will move to sound source in & out of sight, and avoid obstacles • Integrated WebApp with information portraying: location, angle, distance, current direction movement, alert user once destination is reached • Robot movement with periodical listening capability

Thank You,
Any Questions?

Sound Localization Robot 5 – Team 23

Samuel Lackey, Lucas Kettler, Joshua Teele,
Sanjana Pasumarthi